

Human Review: Positive multilinear forms on non-atomic L_p

Original automatic proof: `solution_packet.pdf`

Checked date: 13 June 2026

Status: Manually checked.

Verdict: The proof appears correct.

Human Comments

The proposed solution appears to give a correct negative answer to Question 3 from the source paper, under the natural interpretation that the spaces are in the finite-exponent range $1 < p_i < \infty$. The counterexample is the positive n -linear form

$$A(f_1, \dots, f_n) = \int_0^1 f_1(t) \cdots f_n(t) dt$$

on

$$L_{p_1}[0, 1] \times \cdots \times L_{p_n}[0, 1], \quad \sum_{i=1}^n \frac{1}{p_i} < 1.$$

This form is bounded and positive, but it is not weakly sequentially continuous.

Boundedness and Positivity

The boundedness is the standard Holder argument. Since

$$\sum_{i=1}^n \frac{1}{p_i} < 1,$$

one can choose $r \in (1, \infty]$ such that

$$\sum_{i=1}^n \frac{1}{p_i} + \frac{1}{r} = 1.$$

Applying Holder's inequality to $f_1 \cdots f_n \cdot 1$ gives

$$|A(f_1, \dots, f_n)| \leq \|f_1\|_{p_1} \cdots \|f_n\|_{p_n} \|1\|_r.$$

Thus A is a continuous n -linear form. Positivity is immediate: if each $f_i \geq 0$, then the integrand is non-negative almost everywhere.

Failure of Weak Sequential Continuity

Let (r_k) be the Rademacher functions. For every finite p , the sequence (r_k) is weakly null in $L_p[0, 1]$. Hence $r_k \rightarrow 0$ weakly in each relevant finite L_{p_i} -space.

However,

$$A(r_k, r_k, 1, \dots, 1) = \int_0^1 r_k(t)^2 dt = 1$$

for every k , whereas

$$A(0, 0, 1, \dots, 1) = 0.$$

Therefore A is not weakly sequentially continuous.

Interpretation

The argument is short, but it seems to be a genuine obstruction to the stated question. The main point to check is not the calculation itself, but whether the authors intended the question exactly in this finite-exponent form, or whether some additional hypothesis was meant to exclude this Rademacher-product example.

Review Outcome

Archive this packet as an apparently correct full negative answer, pending author feedback. I recommend contacting the authors to ask whether the argument is known to them, whether it matches the intended formulation of the question, and how they would assess the value of the example.